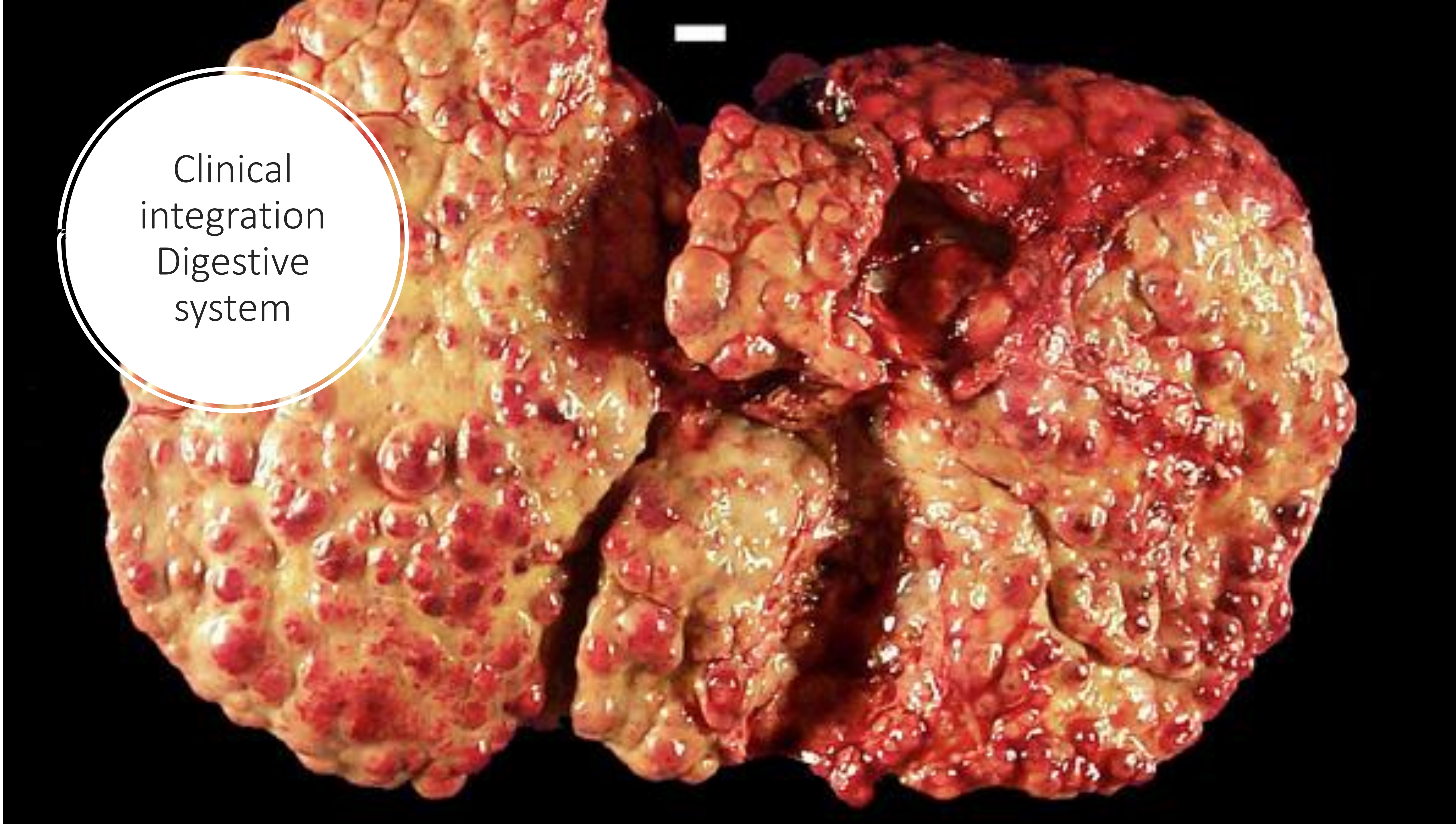
Clinical integration Digestive system



Clinical Case 1

1a. A 42-year-old man is brought to the emergency department because of a 6-day history of severe burning epigastric pain usually occurring 2-3 hours after meals, which frequently wakes him up at night. He has had this type pf pain for the past 3-years but has not sought medical attention because it improved when he took “Pepto-bismol”. The pain is associated with nausea. He has a history of chronic lower back pain for which he takes oral medications regularly. Physical examination shows mild epigastric tenderness and slightly dry mucous membranes. Vital signs show pulse is 89/min and blood pressure is 145/90.

- What are the possible diagnoses?,
- What further information or tests would help make a definitive diagnosis ?
- What risk factors predispose to this condition?

1b. He is admitted to the ward, managed with appropriate medications and discharged after 3 days with oral medications and lifestyle change orientation. Nine months later he returns to the emergency room because of sudden onset of severe abdominal pain and brownish colored vomitus . He admits to being non-compliant with his treatment. Vital signs are pulse is 110/min, blood pressure 150/95 mmHg respiratory rate is 24/min. Physical examination shows severe painful distress, pale mucous membranes, rebound tenderness and guarding on abdominal palpation. Laboratory studies show leukocytosis and anemia.

- What is his current diagnosis?
- What further investigations would help confirm your diagnosis?
- What would the treatment be in this case?

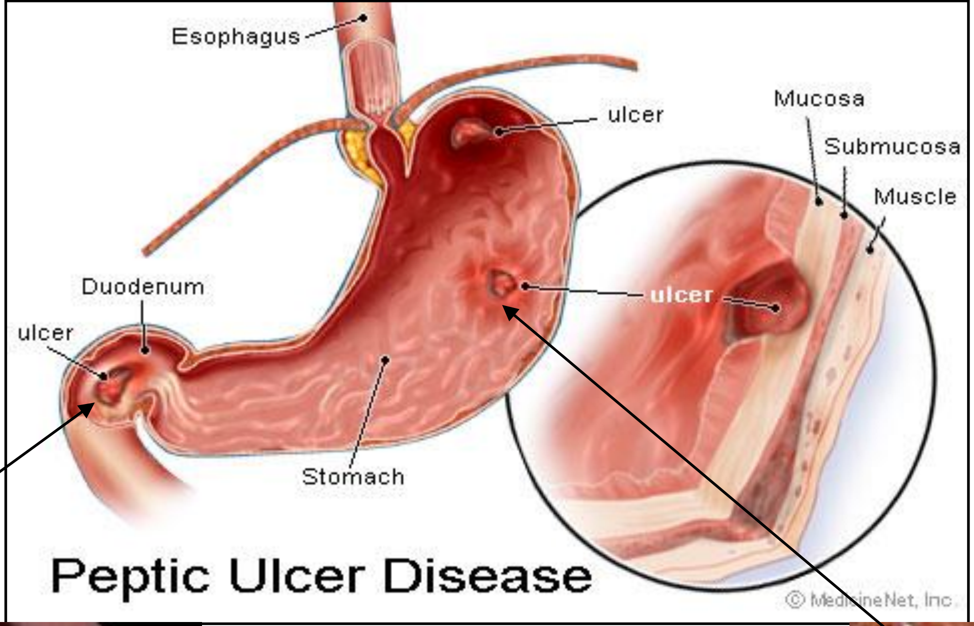
Clinical Case 1

1a. A 62-year-old man is brought to the emergency department because of a 6-day history of severe burning epigastric pain usually occurring 2-3 hours after meals, which frequently wakes him up at night. He has had this type pf pain for the past 3-years but has not sought medical attention because it improved when he took “Pepto-bismol”. The pain is associated with nausea. He has a history of chronic lower back pain for which he takes oral medications regularly. Physical examination shows mild epigastric tenderness and slightly dry mucous membranes. Vital signs show pulse is 89/min and blood pressure is 145/90

- What are the possible diagnoses? Peptic ulcer(duodenal or gastric), Chronic gastritis, Chron’s disease, Gastroesophageal reflux disease, ischemic heart disease, gall-stones
- What further information or tests would help make a definitive diagnosis ? Endoscopy, colonoscopy (r/o Chron’s), biopsy, test for H-pylori→ urease breath test), stool → occult blood, H-Pylori antigen
- What is your definitive diagnosis? Duodenal ulcer→ late onset of pain and alleviation with meals
- What risk factors predispose to this condition? smoking, diet, alcohol, stressful lifestyles, DRUGS: NSAIDS, steroids,
- What treatment was he most likely given? Triple treatment if H-pylori positive(most likely cause) : H-2 blocker, proton pump inhibitor, antibiotics,

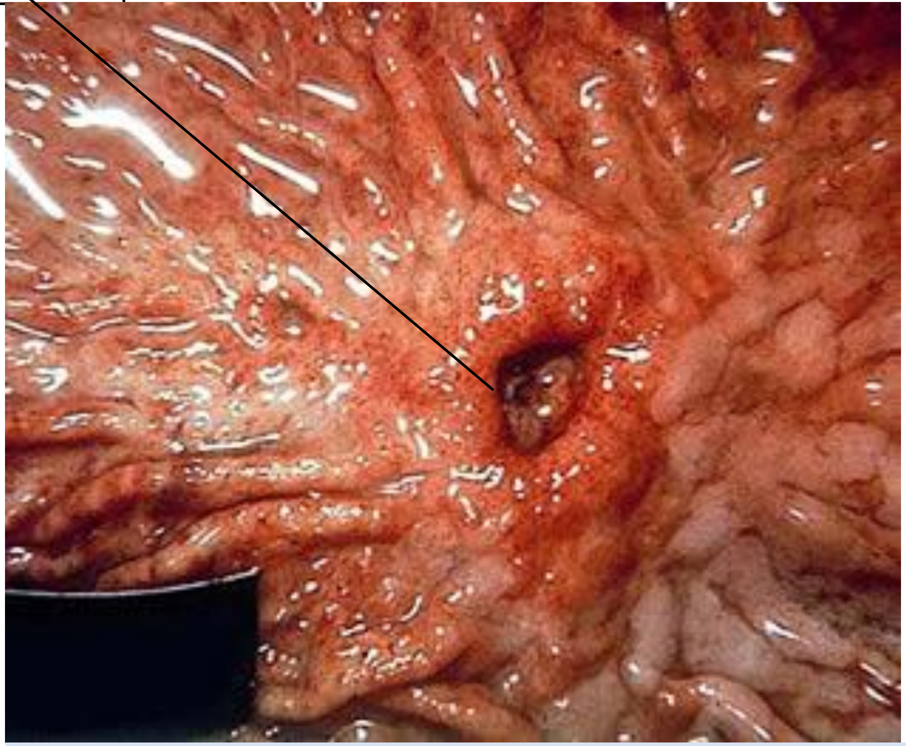
1b. He is admitted to the ward, managed with appropriate medications and discharged after 3 days with oral medications and lifestyle change orientation. Nine months later he returns to the emergency room because of sudden onset of severe abdominal pain and brownish colored vomitus . He admits to being non-compliant with his treatment. Vital signs are pulse is 110/min, blood pressure 150/95 mmHg respiratory rate is 24/min. Physical examination shows severe painful distress, pale mucous membranes, rebound tenderness and guarding on abdominal palpation. Laboratory studies show leukocytosis and anemia.

- What further investigations would help confirm your diagnosis? Radiograph (pneumoperitoneum), less important is ultrasound (fluid in the peritoneum)
- What is his current diagnosis? Perforated duodenal ulcer with chemical peritonitis
- What would the treatment be in this case? Emergency laparotomy with abdominal lavage and ligation of bleeding vessel, repair of organ.

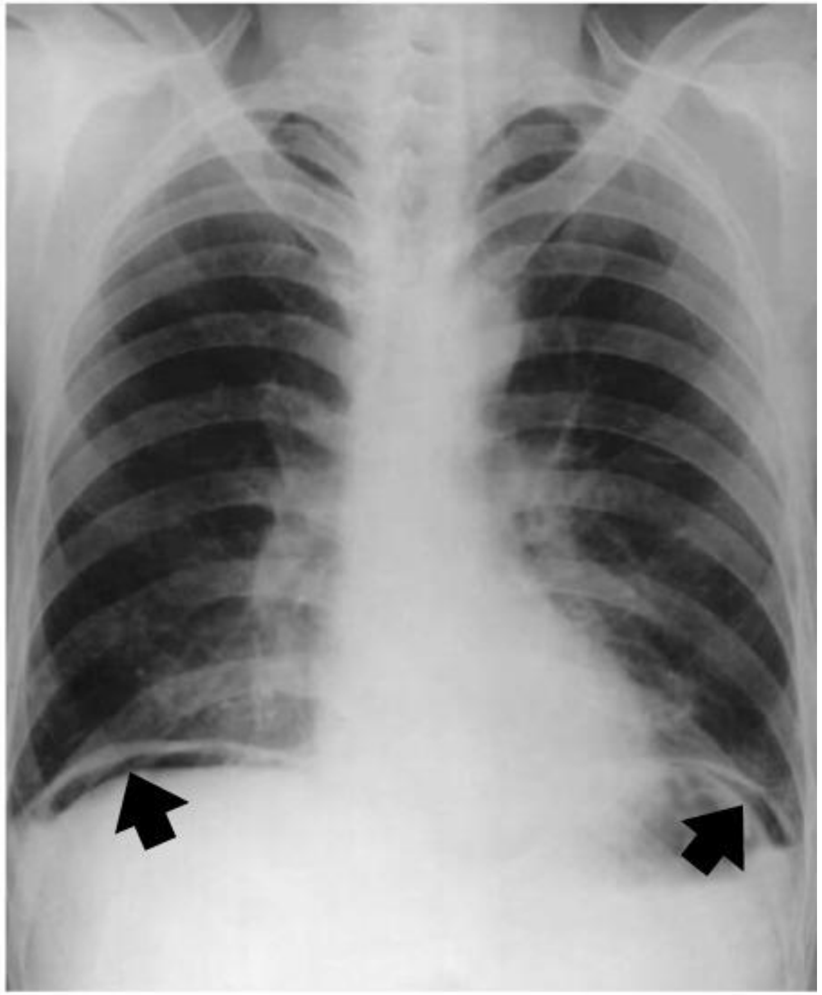


https://www.medscape.com/

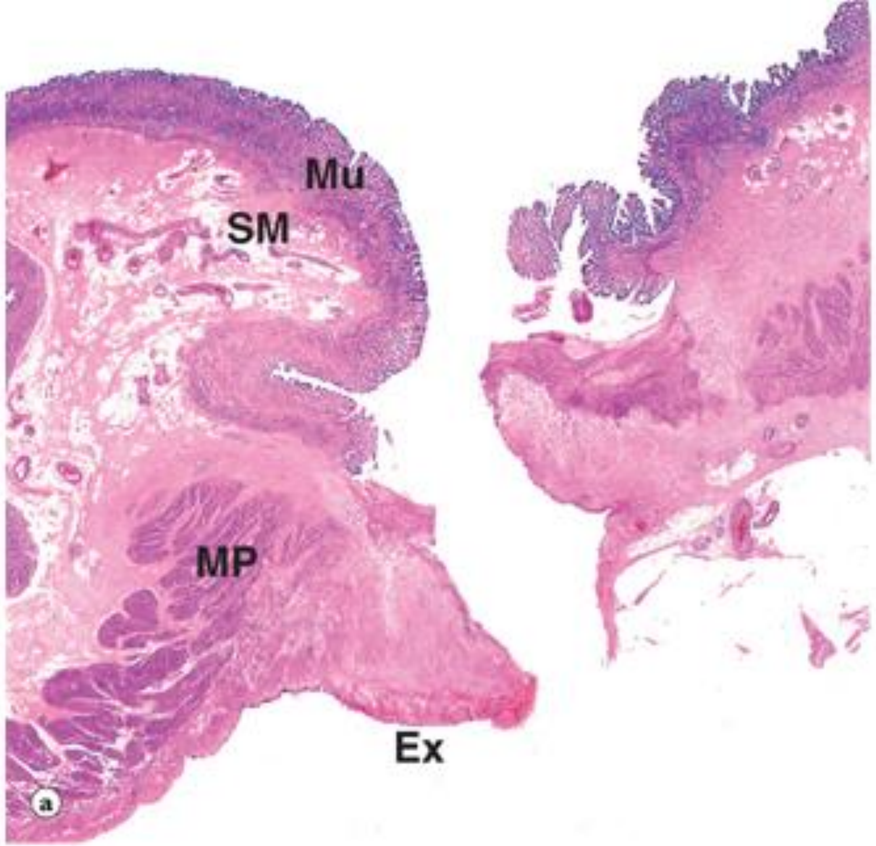
Duodenal ulcer



Gastric ulcer



Air under the diaphragm (pneumoperitoneum) indicates hollow organ perforation



Perforated gastric ulcer. Tissue necrosis has extended through the full thickness of the wall, with complete destruction of the mucosa *Mu*, submucosa *SM* and muscularis *MP*. Peritonitis is manifested by an acute inflammatory exudate *Ex* on the serosal surface of the stomach.

An ulcer can involve mucosa, submucosa and can even involve the muscularis externa and serosa leading to perforation

Tubular contents spill into the abdominal cavity and induce severe inflammation. This condition is called chemical peritonitis.

Clinical Case 2

A 49-year-old man comes to the physician because of weakness, loss of appetite and increasing abdominal girth for the past 6 months. He has a history of chronic alcohol abuse for the past 25 years and had a liver transplantation 3 years ago. He also has had several admissions to the emergency department for alcohol intoxication in the past year. Physical examination shows a firm distended abdomen, gynecomastia, distended veins in the periumbilical region and limbs, mild jaundice and same muscle wasting.

- What are possible diagnoses?
- What further investigations would be required to make a final diagnosis?
- Explain the findings on physical examination.
- Discuss the changes in the liver throughout the evolution of this disease.

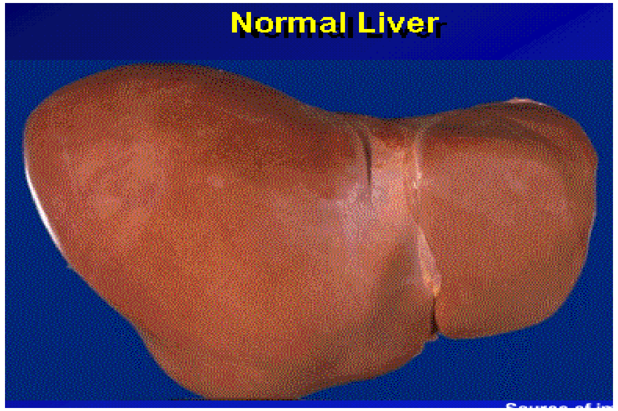
Clinical Case 2

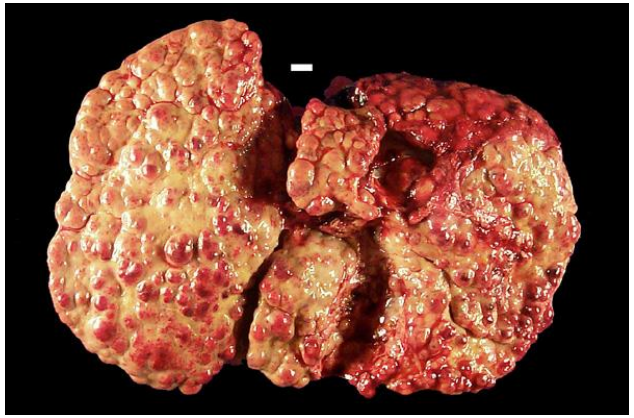
A 49-year-old man comes to the physician because of weakness, loss of appetite and increasing abdominal girth for the past 6 months. He has a history of chronic alcohol abuse for the past 25 years and had a liver transplantation 3 years ago. He also has had several admissions to the emergency department for alcohol intoxication in the past year. Physical examination shows a firm distended abdomen, gynecomastia, distended veins in the periumbilical region and limbs and same muscle wasting.

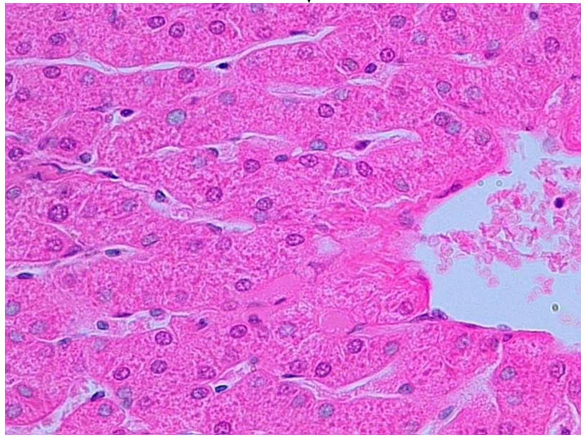
1. What further investigations would be required to make a final diagnosis?
   Ultrasound, CT scan, blood→ liver function
2. Explain the findings on physical examination. Distended abdomen (ascites), gynecomastia (defective hormone metabolism), varicose veins, distended periumbilical veins caput medusae→ see anatomy

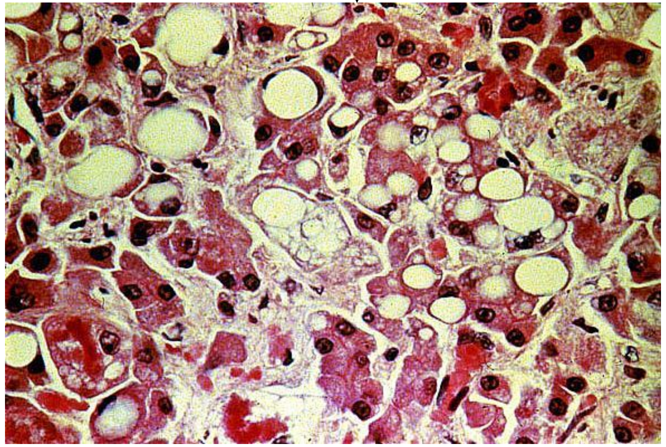
3. Discuss the changes in the liver throughout the evolution of this disease.

- Extensive and chronic parenchymal injury from alcohol abuse leads to loss of liver tissue beyond regenerative capacity.
- Fibrosis will occur → diffuse, extending throughout the liver.
- Fibrosis is irreversible.
- In cirrhosis, the collagen fibers are deposited in all portions of the lobule, accompanied by alterations in the sinusoidal endothelial cells.
- Net result is severe blood flow disruption and decrease in hepatocyte function.
- Nodular appearance is due to extensive fibrosis with areas of normal hepatic parenchyma





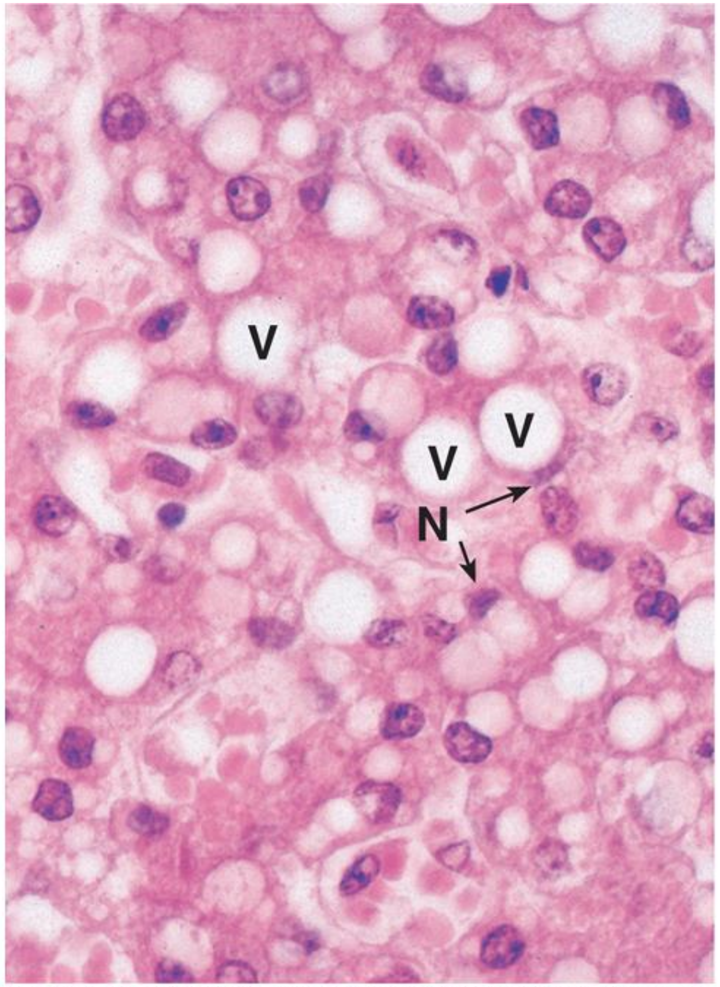




Fatty liver disease in early stages of alcoholic liver disease

Fibrosis resulting in reorganization of the parenchyma in nodules

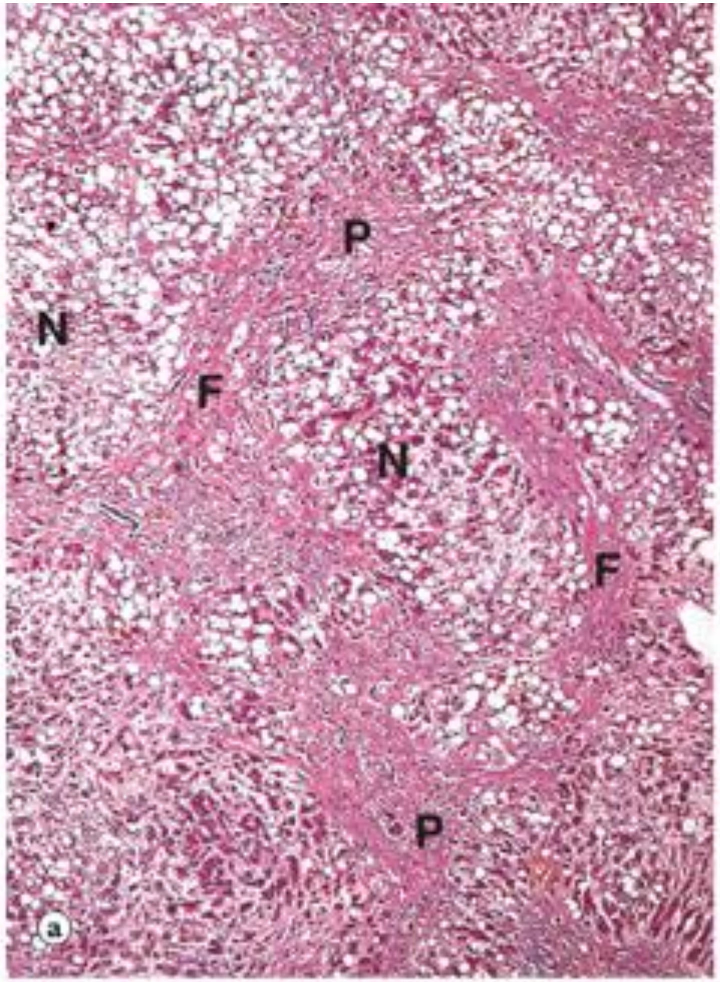
Fatty change in liver



Young et al: Wheater's Basic Pathology: A Text, Atlas and Review of Histopathology, 5th Edition.
Copyright © 2010 by Churchill Livingstone, an imprint of Elsevier, Ltd. All rights reserved.

Liver shows large vacuoles V in the hepatocytes with displacement of the nucleus N.

Alcoholic cirrhosis



Nodular clusters of regenerating hepatic parenchyma (N) interspaced with fibrotic tissue (F). Fibrosis extends to portal area (P)